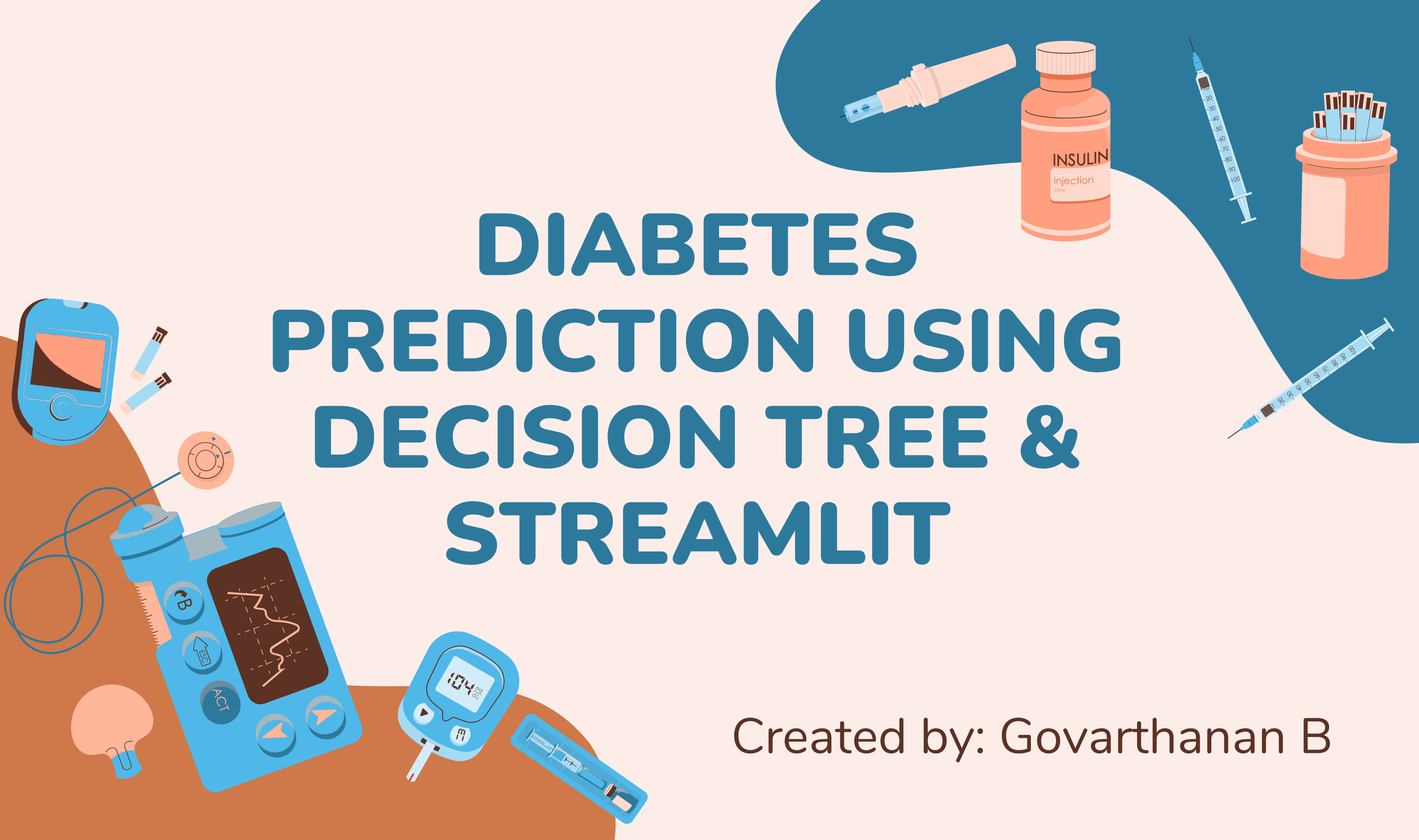


# DIABETES PREDICTION USING DECISION TREE & STREAMLIT

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# WHAT IS DECISION TREE CLASSIFICATION?

- Decision Tree is a supervised machine learning algorithm used for classification and regression.
- It works like a flowchart structure where:
- Each internal node represents a feature
- Each branch represents a decision rule
- Each leaf node represents an output class



# Why Decision Tree for Diabetes Prediction?

## Easy Interpretation:

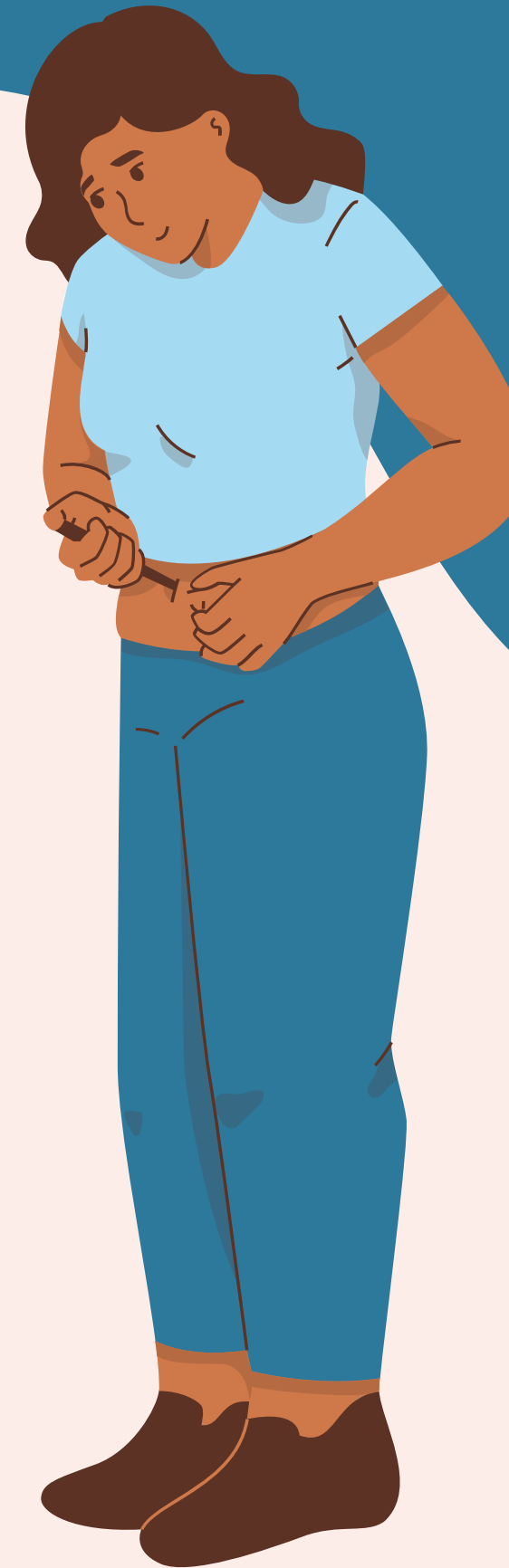
Decision Trees are simple to understand and visualize, making them suitable for medical applications.

## Handles Non-Linear Data:

Medical data often contains complex patterns. Decision Trees can capture these relationships effectively.

## Feature Importance:

The model identifies which medical attributes such as glucose, BMI, or age are important for prediction.



# MODEL INPUTS & OUTPUTS

## Input Features (X)

- Pregnancies
- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- BMI
- Diabetes Pedigree Function
- Age

## Target Output (y)

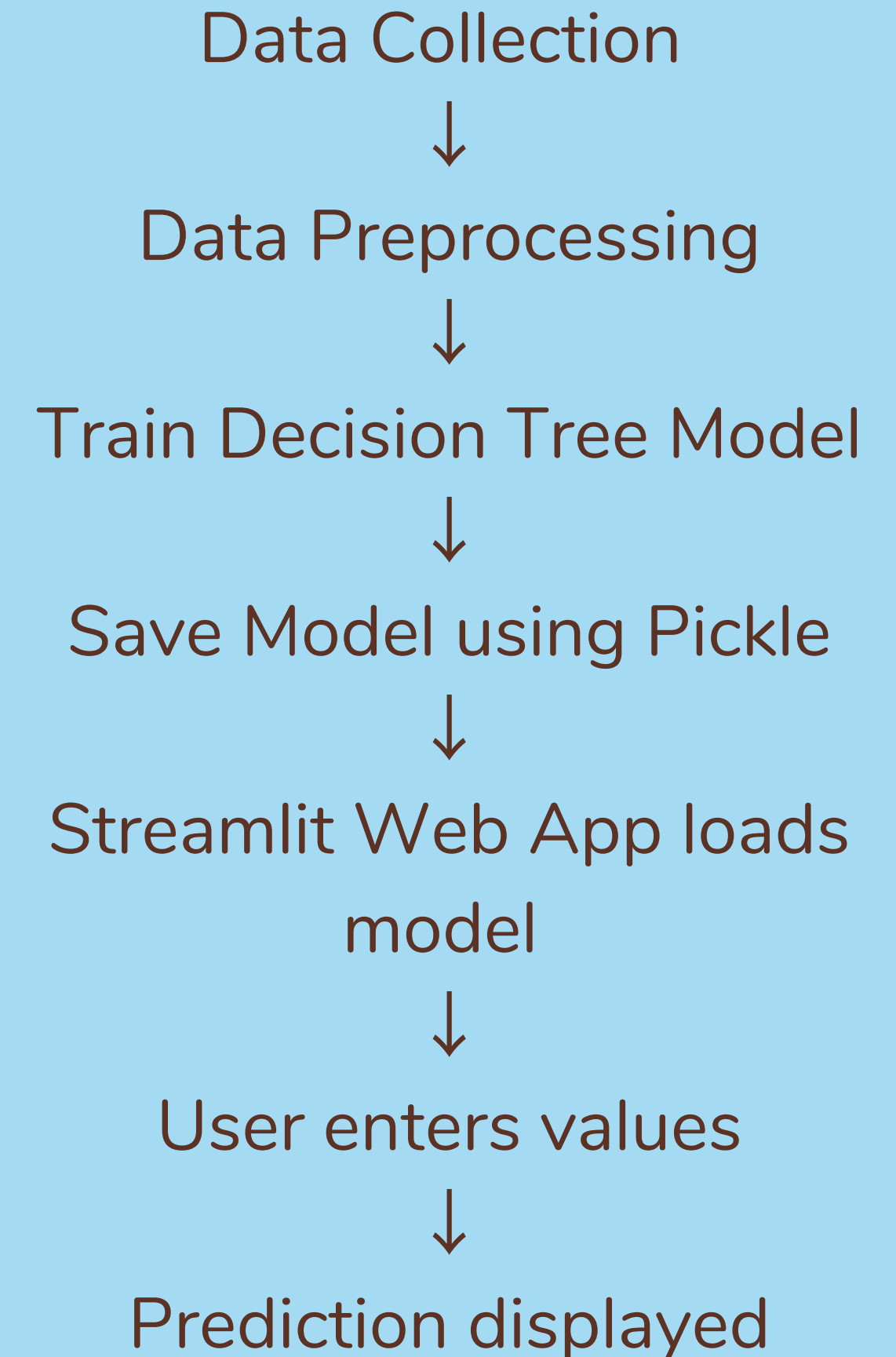
- Outcome
- 0 → No Diabetes
  - 1 → Diabetes

# HOW THE MODEL LEARNS

The model is trained using historical diabetes patient data.

1. Dataset is split into training and testing sets
2. Decision Tree learns patterns between features and outcome
3. Best splits are selected using Gini Index
4. Model predicts whether a person has diabetes

## Model Workflow





# REAL-LIFE APPLICATIONS OF DECISION TREE

## HEALTHCARE :

Treatment recommendation  
systems

## BANKING & FINANCE :

- Loan approval / rejection
- Credit risk assessment

## MANUFACTURING :

- Quality control
- Fault detection in machines

## EDUCATION :

- Student performance  
prediction